

Encoding	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Notes: 8 16-bit registers indexed by a 3-bit address, with separate instruction and data memory. R Type stands for register function work, I Type stands for signed immediate work, J Type stands for jump work, and LI Type stands for load immediate work (bigger numbers if I doesn't work). Op-Code provides 16 distinct commands, and decides instruction type.
R Type	Op-Code			Reg D			Reg S			Reg T			Function				
I Type	Op-Code			Reg D			Reg S			Signed Immediate							
J Type	Op-Code			Address													
LI Type	Op-Code			Reg D			S	Unsigned Byte									

Function	Ops	Effect	Use Cases
000	Add	RD = RS + RT	Adds Registers
001	Sub	RD = RS - RT	Subs Registers
010	And	RD = RS & RT	Masks Bits
011	Or	RD = RS RT	Merges Bits
100	Xor	RD = RS ^ RT	Flips Bits / Compares Registers
101	Not	RD = ~RS	Inverts Register
110	Shift L	RD = RS << RT	Doubles RS RT Number of Times
111	Shift R	RD = RS >> RT	Halves RS RT Number of Times

Op-Code	Ops	Format	Effect	Use Cases
0000	R RD, RS, RT, Function	R	Function Effect	See Function Table
0001	LI RD, Hi/Lo, Byte	LI	Load Byte into Half of RD	Adds Byte into High/Low Byte of RD, based on Hi/Lo S bit
0010	Addi RD, RS, Imm	I	RD = RS + Imm (Signed, ± 32)	Adds Signed 6-bit Imm into RS for counters ($i = i + 1$)
0011	LW RD, Imm(RS)	I	RD = dataMem[RS + Imm]	Locates dataMem Reg at base RS + offset Imm and loads register into RD
0100	SW RD, Imm(RS)	I	dataMem[RS + Imm] = RD	Locates dataMem Reg at base RS + offset Imm and loads RD into register
0101	BEQ RD, RS, Offset	I	If RD == RS, PC += Offset	Branches to offset PC value if RD = RS
0110	BNE RD, RS, Offset	I	If RD != RS, PC += Offset	Branches to offset PC value if RD != RS
0111	BLT RD, RS, Offset	I	If RD < RS, PC += Offset	Branches to offset PC value if RD < RS
1000	J Address	J	PC = Address	Jumps to Address PC value

Port Map				
Module	Port	Dir	Width	Type
ALU	a	in	16	wire
	b	in	16	wire
	op	in	3	wire
	result	out	16	reg
	zero	out	1	wire
Sign Extender	sign	out	1	wire
	in	in	6	wire
Register File	out	out	16	wire
	addr1 / addr2 / addrw	in	3 each	wire
	dataw	in	16	wire
	enablew / clk / rst	in	1 each	wire
PC	out1 / out2	out	16 each	wire
	offset	in	16	wire
	jaddr	in	12	wire
	pcsel	in	2	wire
	clk / rst	in	1 each	wire
	pcout	out	16	reg
	instr	out	16	wire
Instruction Memory	addr	in	8	wire
Data Memory	instr	out	16	wire
	addr	in	16	wire
	dataw	in	16	wire
	enablew / clk	in	1 each	wire
	dataout	out	16	wire